

VERKHNY UFALEY, Russian Federation

WMO: 285410

Lat: 56.083N

Lon: 60.300E

Elev: 399

StdP: 96.63

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-27.3	-33.5	0.2	-30.5	-30.2	0.2	-27.1	7.9	-5.5	7.2	-8.7	0.8	40	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	28.6	18.5	26.7	17.7	25.0	16.8	19.8	26.2	18.9	24.8	17.9	23.3	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.2	22.5	16.7	12.5	21.5	15.8	11.8	20.5	58.4	26.3	55.3	24.6	52.2	23.3	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.3	5.5	DB	-34.1	31.3	3.4	1.8	-36.6	32.6	-38.6	33.6	-40.5	34.6	-43.0	35.9	(4)
(5)			WB	-34.2	21.2	3.4	0.9	-36.7	21.8	-38.7	22.3	-40.6	22.8	-43.1	23.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-13.9	-12.1	-5.0	3.3	10.4	15.0	17.0	15.1	9.3	2.6	-6.1	-11.8	(6)
(7)		DBStd	12.26	6.72	6.80	4.87	5.15	4.95	4.36	3.65	4.11	4.35	4.90	6.88	7.15	(7)
(8)		HDD10.0	3561	739	619	465	206	58	9	1	5	64	232	484	677	(8)
(9)		HDD18.3	6002	998	853	723	450	249	116	68	118	272	487	734	936	(9)
(10)		CDD10.0	663	0	0	0	7	69	159	219	162	43	4	0	0	(10)
(11)		CDD18.3	64	0	0	0	0	3	16	27	17	1	0	0	0	(11)
(12)		CDH23.3	781	0	0	0	2	60	208	301	191	20	0	0	0	(12)
(13)		CDH26.7	169	0	0	0	0	4	44	66	52	3	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.9	2.1	2.5	2.4	2.3	2.1	1.7	1.7	1.9	2.4	2.3	2.1	(14)
(15)	Precipitation	PrecAvg	561	27	20	27	35	54	71	91	68	44	52	40	32	(15)
(16)		PrecMax	790	62	47	51	94	138	134	207	135	103	119	73	69	(16)
(17)		PrecMin	368	6	6	4	3	11	18	32	14	7	5	6	8	(17)
(18)		PrecStd	108	13	9	15	22	34	30	43	33	21	27	18	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	2.6	9.6	22.0	27.1	30.2	31.1	31.0	25.9	18.2	8.1	1.6	(19)
MCWB			0.4	0.4	4.1	12.0	15.0	18.4	19.1	18.7	15.1	10.4	5.4	0.1	(20)	
2%		DB	-1.2	0.6	5.8	17.8	25.1	27.9	28.4	27.9	22.3	15.0	5.5	0.1	(21)	
		MCWB	-2.1	-1.2	2.0	9.3	14.3	18.0	18.9	18.7	14.3	9.2	4.0	-1.0	(22)	
5%		DB	-2.8	-1.0	3.5	14.8	23.0	25.8	26.5	25.2	19.2	12.2	3.4	-0.9	(23)	
		MCWB	-3.5	-2.4	0.6	8.1	13.7	17.1	18.4	17.9	13.2	7.9	2.0	-1.8	(24)	
10%		DB	-4.6	-2.6	1.8	11.8	20.4	23.6	24.7	22.7	16.6	9.7	1.6	-2.5	(25)	
		MCWB	-5.4	-3.9	-0.4	6.4	12.2	16.2	17.6	16.9	11.9	6.9	0.5	-3.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.0	4.5	12.3	17.3	20.5	21.0	20.6	17.2	11.6	5.8	0.3	(27)
MCDWB			1.3	2.4	8.2	20.9	23.4	27.3	27.0	27.3	23.1	16.3	7.0	1.1	(28)	
2%		WB	-2.0	-0.9	2.4	10.2	15.3	19.1	20.0	19.5	15.1	9.7	4.1	-0.8	(29)	
		MCDWB	-1.3	0.3	5.0	16.0	22.7	25.5	26.1	25.9	20.8	13.8	5.4	0.0	(30)	
5%		WB	-3.5	-2.3	1.0	8.5	14.1	17.9	19.1	18.3	13.7	8.3	2.1	-1.8	(31)	
		MCDWB	-2.8	-1.1	2.9	14.0	21.3	23.9	24.9	23.8	18.1	11.5	3.1	-0.9	(32)	
10%		WB	-5.3	-3.9	-0.2	6.9	12.9	16.8	18.2	17.2	12.4	6.9	0.7	-3.3	(33)	
		MCDWB	-4.7	-2.6	1.5	11.4	19.0	22.4	23.4	21.9	16.0	9.5	1.5	-2.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	9.1	9.2	10.0	12.0	11.0	10.6	9.9	9.0	6.2	5.7	6.4	(35)
MCDBR			5.8	9.0	11.4	15.2	16.1	14.8	14.4	14.7	13.6	12.4	6.8	5.7	(36)	
MCWBR			5.3	7.3	8.1	8.1	7.5	6.4	6.2	6.5	7.0	7.2	5.5	5.2	(37)	
5% WB		MCDBR	5.6	7.5	8.9	13.4	14.2	12.9	12.3	12.8	11.9	10.9	5.6	5.7	(38)	
		MCWBR	5.2	6.3	6.6	7.6	7.3	6.2	5.8	6.2	6.7	6.8	5.0	5.3	(39)	
		(40)	Clear-Sky Solar Irradiance	taub	0.277	0.304	0.359	0.402	0.402	0.396	0.427	0.437	0.378	0.345	0.302	0.274
taud	2.070	2.058		1.977	2.105	2.241	2.327	2.233	2.175	2.313	2.275	2.159	2.034	(41)		
Ebn at Noon	596	734		776	808	837	847	808	767	768	690	563	486	(42)		
Edh at Noon	62	94		132	136	126	118	127	126	96	77	57	50	(43)		
(44)	All-Sky Solar Radiation	RadAvg	0.75	1.76	3.20	4.22	5.07	5.50	5.35	4.13	2.69	1.30	0.64	0.45	(44)	
RadStd		0.08	0.14	0.25	0.48	0.53	0.51	0.42	0.42	0.21	0.25	0.08	0.07	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page